

Rashi Goyal

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Professional Summary

Associate AI Engineer with a Master's from **IIT Kharagpur**, specializing in AI/ML and production-grade software architectures. Backed by professional experience building end-to-end AI systems, deploying cloud-native architectures (AWS Bedrock), and optimizing relational/vector database pipelines. Two-time **National Hackathon Winner** with strong foundations in Data Structures and Algorithms (C++) and international research exposure. Proven ability to deliver scalable, client-ready software within fast-paced Agile environments.

Education

Indian Institute of Technology, Kharagpur (IIT Kharagpur) **2020 – 2025**
Integrated (Master's + Bachelor's) in Geophysics *CGPA: 8.36 / 10.0*
Micro-Specialization: Artificial Intelligence & Machine Learning (NLP, Deep Learning, Statistics for ML)

Technical Skills & Certifications

- **AI Tools:** Claude, Cursor, Windsurf, Antigravity, Copilot
- **Languages & Core:** Python, C++, JavaScript, SQL, OOP, Data Structures & Algorithms.
- **Frameworks & Cloud:** React, React Native, .NET (C#), Node.js, AWS (S3, Lambda, API Gateway), Docker, Redis
- **Methodologies & Tools:** Agile Scrum, Git, RESTful API Design (3-Layer Architecture).

Work Experience

Associate AI Engineer | Cambridge Technology, Hyderabad **Jun 2025 – Present**

- Designed the validation workflow to combine **deterministic rules and AI-based semantic comparisons**, using code for structured checks and AI models for ambiguous product and content comparisons in Digital Shelf Compliance.
- Automated large-scale validation using **Playwright**, replacing a manual process that would take significant time to validate hundreds of products across multiple retailers; AI-assisted analysis to review validation accuracy.
- Architected and deployed end-to-end **agentic AI systems** using **Amazon Bedrock**, implementing custom guardrails, prompt workflows, and real-time API-driven tool invocations; evaluated system behavior using **golden query datasets**.
- Built cloud-native data and API workflows using **AWS Lambda, S3, API Gateway, and CloudFront**, supporting scalable data processing and application delivery.

Software Engineering Intern (GenAI Systems) | FutureSmartAI **Mar 2024 – May 2024**

- Built end-to-end production pipelines for enterprise document intelligence utilizing **LangChain** and **LlamaIndex**.
- Designed an optimized **Retrieval-Augmented Generation (RAG)** pipeline, configuring a **Chroma Vector Database** to execute highly efficient, low-latency semantic search queries.

Research Intern (Mitacs Globalink) | McMaster University, Canada **May 2023 – Jul 2023**

- Developed scalable **Python** pipelines for geophysical predictive modeling and rigorous data validation.
- Automated complex analysis workflows to ensure strict reproducibility and system scalability across large datasets.

Software Developer | SkilledSpeech (Remote) **Jan 2026 – Jun 2026**

- Spearheaded development of a **React Native** mobile application, managing cross-platform frontend-to-backend.
- Engineered background service communication logic by integrating push/email notification architectures via **Fire-base/GCP** and worked on various other features.

Awards & Achievements

- **National Winner** – Dark Pattern Buster Hackathon (2023): Engineered an end-to-end Chrome extension featuring a fine-tuned **RoBERTa** classification with real-time scoring capabilities, detecting dark patterns on ecommerce websites.
- **National Winner** – Smart India Hackathon (2023): Built an encoder-decoder **LSTM model** for vessel trajectory predictions and estimation of oil usage, decreasing simulated collision tracking margins by up to 30%.
- **Grand Finalist** – Google Girl's Hackathon (2024): Developed a multi-factor top 3 medical recommendation doctors application based on illness described in simple english leveraging TF-IDF and Cosine Similarity, hosted on AWS.
- **KL Chopra Award Nominee** – Best Master's Thesis: Authored a novel lunar crater detection paradigm using automated pipeline tuning with a **YOLOv11** backbone by creating bounding boxes on images to classify.
- **Amazon ML Summer School (2023)**: Selected to participate among the top 0.05% of applicants nationwide.